

**Comprehensive Analysis of Formula 1 World Championship Trends
and Performance Metrics**

Prepared by

Dhyey Kasundra, Riyazul Ansari, Sanyami Naik, Sakshi Deshmukh

IS 525: Data Warehousing and Business Intelligence

Guided by Prof. Michael Wonderlich

May 12th, 2025

Project Scenario

Every millisecond, pit stop, and strategy decision can affect the result in Formula 1 (F1), one of the most data-rich sports in the world. The sport has gathered a vast collection of race data, driver statistics, constructor performance, and seasonal trends over the years. With so much data available, the opportunity to transform raw figures into meaningful insights becomes not only valuable but essential for enthusiasts, analysts, and strategic decision-makers. This project aims to identify important trends in the competitive ecosystem of the F1 championship by utilizing interactive dashboards and descriptive analytics. It aims to transform raw data into actionable insights that improve F1 viewership.

The primary objective of this project is to explore and visualize past Formula 1 statistics, emphasizing how various elements such as constructor dominance, driver contributions, and seasonal fluctuations influence the performance environment. Race results, constructor standings, and driver statistics from several seasons were cleaned and organized by the team to create extensive Tableau dashboards that present performance trends in an accessible and engaging form. Here, the focus is on giving clarity through visual storytelling and establishing high-impact measures that characterize success in the sport, rather than predictive modeling.

F1 analysts, team strategists, fantasy league participants, sports journalists, and passionate followers who are eager to learn more about the sport's development are part of the project's target audience. The visualizations are intended to provide these users with easy-to-use tools for analyzing race outcomes, monitoring team performance, and understanding the complex relationships between constructor and driver performance. By integrating qualifying trends and circuit-level race

behavior, the dashboards deliver richer, multi-dimensional insights for both tactical and historical exploration.

What is the Problem You Are Trying to Solve?

The project aims to uncover trends, performance patterns, and strategic insights in Formula 1 racing by visualizing key aspects of the sport across four dimensions: individual driver performance, team success, circuit dynamics, and qualifying metrics.

By bringing together multiple datasets from historical F1 races, the project solves the problem of fragmented or hard-to-interpret performance data. It enables users to answer questions like:

- How have top drivers and teams evolved over time?
- Which circuits are most iconic or frequently used?
- What are the average qualifying positions?
- What are the dominant nationalities in F1 performance?

The dashboards convert complex Formula-1 data into intuitive, interactive visuals that support performance analysis, fan engagement, and strategic exploration.

Where did your data come from?

The data for this project was sourced from the Formula 1 World Championship dataset published by Rohan Rao on [Kaggle](#). This dataset compiles comprehensive historical Formula 1 data from 1950 to 2024, originally extracted and curated from the official Ergast Developer API, a well-known and regularly updated public database of F1 race results and statistics.

Who is the Intended Audience?

The dashboards are designed to serve a wide spectrum of users, ranging from casual enthusiasts to domain experts, each benefiting in different ways:

- Formula 1 Fans and Enthusiasts
 - Gain visual, interactive access to career trends of favorite drivers and teams.
 - Explore season-by-season performance, team dynamics, and circuit usage.
 - Understand how elements like pit stops or qualifying position affect outcomes.
- Sports Analysts and Data Scientists
 - Use the visualizations to detect patterns, anomalies, and performance shifts over time.
 - Perform comparative analysis across drivers, teams, seasons, and track types.
 - Integrate insights into larger predictive models or reporting workflows.
- Journalists and Broadcasters
 - Quickly extract storylines (e.g., Verstappen vs Hamilton seasons, constructor dominance).
 - Use visual evidence to support commentary and enrich pre/post-race coverage.
 - Leverage clean data visuals for audience-friendly broadcast infographics.

- Teams, Strategists, and Motorsport Professionals (hypothetical use)
 - Analyze pit stop performance and timing strategies.
 - Evaluate how specific drivers contribute to the constructor points.
 - Identify high-performing circuits or ideal conditions for success.
- Students, Educators, and Researchers
 - Learn principles of data visualization and ETL through a real-world dataset.
 - Understand how raw motorsport data can be transformed into actionable insights.
 - Use the dashboards as a case study in courses related to analytics, sports management, or data storytelling.

Please provide contact information for your client.

Feedback

I found the dashboards the team presented to be a compelling and insightful summary of Formula 1 trends in performance. Most striking was the elegance in visual storytelling—each visualization served a clear function and reacted interactively to filters, where it was extremely easy to generate insights across circuits, drivers, and seasons. Presenting KPIs alongside granular driver and constructor-level charts meant both high-level and detailed explorations were feasible with Filters: Year range, Nationality, Constructor. The Dashboard features included tooltips, diverging color schemes, and multi-filter interactivity. The balance between simplicity and richness of data was well met. I particularly enjoyed the fact that the dashboards would be useful for both casual and strategic users. With a few minor additions—such as including driver comparison toggles or summary takeaways—the dashboards would be excellent tools for educational, analytical, or even business F1 uses.

Dashboards overview:

1. Driver Performance over seasons(2000-2024) Dashboard overview

Data Integration and Preparation

This dashboard was sourced from 6 separate CSV files: drivers.csv, drivers_standings.csv, lap_times.csv, qualifying.csv, races.csv, and results.csv. Data preparation and transformation (ETL) were performed using Tableau Prep Builder.

Key steps included:

1. Renaming columns for consistency and clarity.
2. Removing redundant or irrelevant columns.
3. Handling null values appropriately (e.g., removing rows with missing IDs or race outcomes).
4. Joining the datasets into a single unified table on relevant keys like raceId and driverId.

The processed output was exported as a new CSV file and used as the sole data source in Tableau Desktop for building the visualizations.



Analysis of the dashboard

The dashboard comprises four main visualizations to present a comprehensive picture of the driver's performance over the last 25 years:

- 1. Drivers' Points Over the Season:** A line chart showing how driver points accumulate across seasons. This view highlights performance consistency and year-over-year improvement. The chart is interactive and dynamically updates when a user selects a different driver from the **Full Name** dropdown filter at the top right.
- 2. Qualifying Position vs Final Position:** A scatter plot comparing drivers' qualifying positions against final race results. Color coding indicates whether a driver improved, dropped, or maintained their position, using a PositionChange legend. The chart updates based on the driver selected in the same dropdown filter.
- 3. Fastest Lap by a Driver in Every Race:** A bar chart that displays the fastest lap (in seconds) achieved by the selected driver in each race, showing trends in race pace across seasons. Like the previous two, this chart also responds to changes in the driver filter.

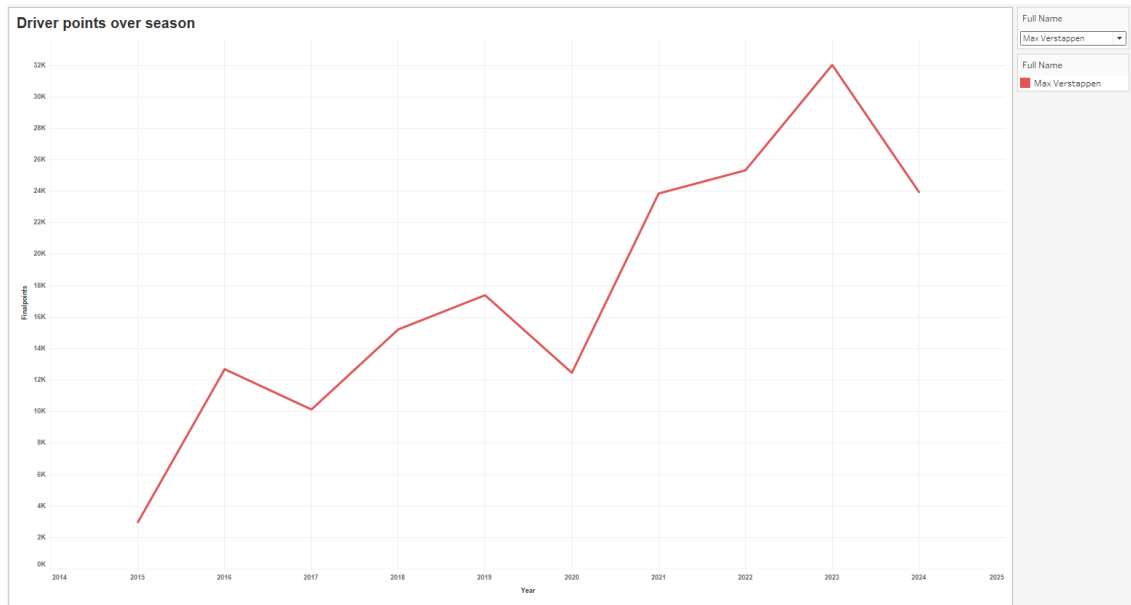
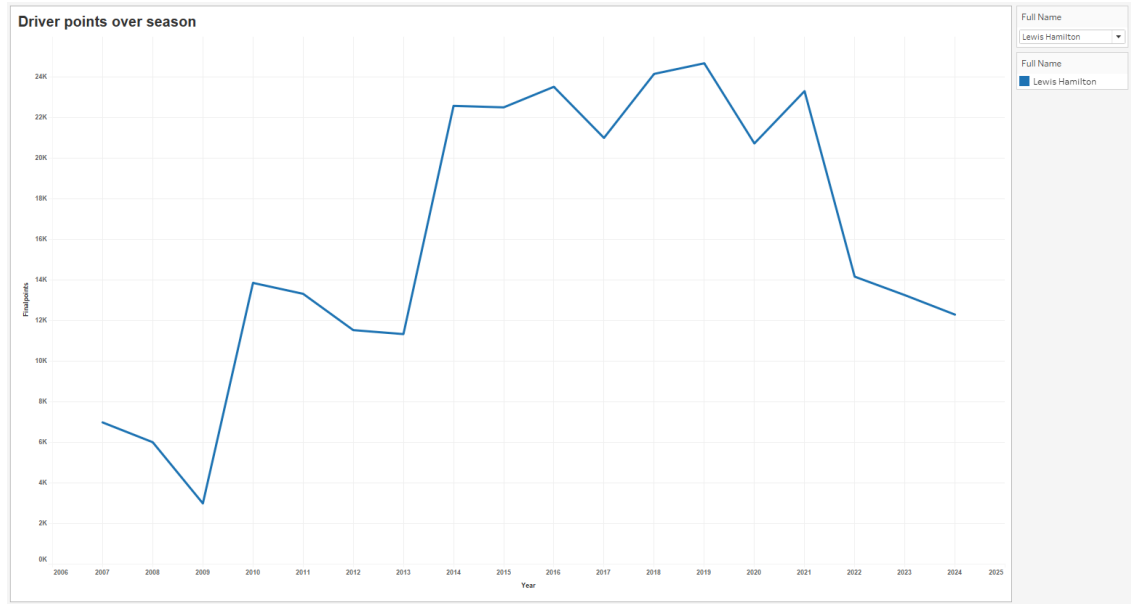
4. **Total Points by Driver Grouped by Nationality:** A treemap visualization comparing drivers' total points, grouped by their nationality. This is controlled by two interactive filters: a **Nationality** dropdown and a **Year Range** slider. These filters allow users to explore point distributions across countries and time periods, offering geographic and historical context.

Discoveries from visualization

1. Max Verstappen's 2021 season clearly stood out with consistent point scoring and fast laps.
2. Drivers like Lewis Hamilton displayed a high number of position gains, often outperforming their qualifying position.
3. The treemap revealed that a few nationalities dominate point accumulation, especially from countries like the UK, Germany, and the Netherlands.
4. Some drivers consistently posted fast laps but didn't always finish high, suggesting high pace but perhaps race-day inconsistency or mechanical issues.
5. The time span from 2000 to 2024 allows us to observe the rise and fall of dominant drivers, shifts in team performance, and changing competitiveness of the grid.
6. A comparative look at the dashboards for Lewis Hamilton and Max Verstappen reveals notable trends:
 - a. Hamilton demonstrates sustained performance with a major leap in points starting in 2014, peaking between 2017–2021 before a gradual decline after

2021. His consistency in fastest laps, especially at recurring circuits like Belgian and Singapore GPs, reflects both adaptability and elite pace over time.

- b. His qualifying vs final position chart shows a mix of dropped and improved positions, with a visible frequency of blue lines (improvements), affirming strong in-race overtaking skills. Notable improvements are seen even from mid-grid starts.
- c. Verstappen, on the other hand, shows a clear acceleration in form starting from 2020, with steep point increases through to 2023. The 2021–2023 phase marks his championship-level performance, with the highest point accumulation and consistently strong fastest laps.
- d. His fastest laps are often recorded at high-speed circuits like the Belgian GP, indicating car performance optimization and high-speed consistency.
- e. The qualifying vs final position chart for Verstappen shows many races where he either maintained or improved upon strong qualifying positions (lots of orange and green), highlighting his ability to convert front-row starts into race wins more often than not.
- f. This comparison illustrates the evolving competitive narrative between a seasoned champion and a rising dominant force in the sport.



7. The Treemap visualization adds additional layers of insight:
 - a. The United Kingdom dominates in total driver points, primarily due to Lewis Hamilton's overwhelming contribution, supported by Jenson Button, Lando Norris, and George Russell.

- b. Germany follows closely, showcasing not just Sebastian Vettel and Michael Schumacher, but also consistent support from Nico Rosberg and Nico Hülkenberg, reflecting depth over decades.
- c. The Netherlands stands out through Max Verstappen alone, indicating how one elite driver can carry national point representation.
- d. Countries like Finland and Spain display balanced contributions, with Raikkonen and Bottas for Finland, and Alonso and Sainz for Spain.
- e. The treemap effectively visualizes how some nations have broad driver depth while others concentrate performance through standout individuals.
- f. It also supports temporal analysis through the year slider, enabling identification of rising drivers and shifts in dominance by nationality over time.

Challenges

1. Managing and transforming large datasets (e.g., lap_times had over 400,000 records).
2. Handling missing or malformed fastestLapTime data.
3. Ensuring joins in Tableau Prep maintained row integrity and did not duplicate data.
4. Initial attempts at using fastestLapTime (in time format) failed in Tableau and had to be replaced with milliseconds-based calculations.
5. Selecting the correct data sources: The original dataset contained 14 CSV files, from which only 6 were used. Choosing the right ones required a deep understanding of their relevance to the dashboard objectives, while ensuring

that no important data was accidentally excluded. This step was essential to build a clean, performant, and accurate visualization base.

Any adjustments you had to make from the original plan?

1. Planned to use fastestLapTime in time format but switched to using milliseconds due to datatype issues.
2. Switched final visualization from “Wins per Season” to a treemap view of “Total Points by Nationality” for better space usage and interpretability.
3. Simplified some visualizations by converting aggregated fields into dimensions to avoid unwanted summing.
4. Originally intended to allow multi-driver comparisons in the visualizations by enabling multi-select in the Full Name filter. However, this created cluttered and less interpretable visuals. Therefore, the approach was revised to show data for a single driver at a time for the first three visualizations (line chart, scatter plot, and bar chart), improving clarity and user experience.

2. Constructor (Team) Success Dashboard Overview:

The dashboard development process involved several key phases to create an interactive visualization of the team's success journey.

Data Integration and Preparation

To build the dashboard, the raw data required a thorough preprocessing pipeline. This was handled using Python’s powerful data preprocessing libraries, Pandas and NumPy. During preprocessing, null-like values denoted by “\N” were standardized and replaced with "Unknown", date and time columns were converted into human-readable formats with data type “datetime”, and redundant or irrelevant fields were dropped to streamline analysis. Duplicate rows were also eliminated to maintain data integrity.

Six core dataset files were utilized to power the dashboard: constructors.csv, drivers.csv, races.csv, constructor_standings.csv, results.csv, and status.csv.

The screenshot displays the Tableau interface. On the left, the 'Connections' pane shows a list of files: constructor_standings.csv, constructors.csv, drivers.csv, races.csv, results.csv, and status.csv. The 'results.csv' file is selected. In the center, a diagram shows the relationships between these files, with 'results.csv' as the central hub connected to the other five files. On the right, the 'Fields' pane shows a list of fields: Result Id, Race Id, Driver Id, Constructor Id, Number, Grid, Position, and Position Text. Below this, a sample data table is shown with 7 rows and 8 columns.

Result Id	Race Id	Driver Id	Constructor Id	Number	Grid	Position	Position Text
1	18	1	1	22	1	1	1
2	18	2	2	3	5	2	2
3	18	3	3	7	7	3	3
4	18	4	4	5	11	4	4
5	18	5	1	23	3	5	5
6	18	6	3	8	13	6	6
7	18	7	5	14	17	7	7

All datasets were joined and loaded directly in Tableau, using the visual data model interface. As shown in the attached diagram, the relationships were constructed carefully based on primary keys like raceId, driverId, and constructorId, ensuring a robust foundation for the dashboard visuals.

Analysis of the Dashboard

The dashboard is made up of four visual charts and a central KPI section, designed to present a comprehensive picture of a constructor's legacy in F1 history.

1. **KPIs (Key Performance Indicators):** Displays the selected constructor name along with three performance indicators: total wins, total points, and the best performing driver (based on their contribution to team points). This area gives users an instant performance snapshot, and the values update dynamically based on the selected constructor.

For example, selecting Mercedes might highlight Lewis Hamilton as their best-performing driver based on the driver's contribution in winning the race, reinforcing his legendary status.

2. **Points & Wins per Season:** Located in the Top Left corner of the dashboard. A dual-coded bar chart showing total points and number of wins (height of bar) and win occurrence (bar color) per season. Bars are color-coded to indicate winning seasons, making it easy to identify dominant years at a glance. This helps users track year-over-year team performance and identify peak periods.

For example, Mercedes' peak in 2016 with 765 points and 208 wins reflects one of the most dominant seasons in modern F1 history.

3. **Driver Contributions to Team Points:** Located in the Top Right corner of the dashboard. Represented as a donut chart, this visual illustrates the percentage contribution of each driver to the constructor's total points. It effectively highlights driver impact and team dependency.

For example, as seen in the case of Mercedes, drivers like Lewis Hamilton had notably large shares with 50.55% of contribution in the total team points, showing their era of influence.

4. **Constructor Rankings Over Time:** Located in the Bottom Left corner of the dashboard. A line chart plotting the team's overall championship position season by season. This shows the rise and fall in competitiveness across years. It highlights trends of rise, dominance, or decline over time.

For example, Mercedes's surge from mid-table to frequent top-4 rankings after 2022 reflects their strategic transformation and competitive edge.

5. **Top 10 Most Successful Teams:** Located in the Bottom Right corner of the dashboard. A horizontal bar chart ranks the top constructors of all time based on total race wins, providing a broader historical context. It demonstrates the macro view of long-term success.

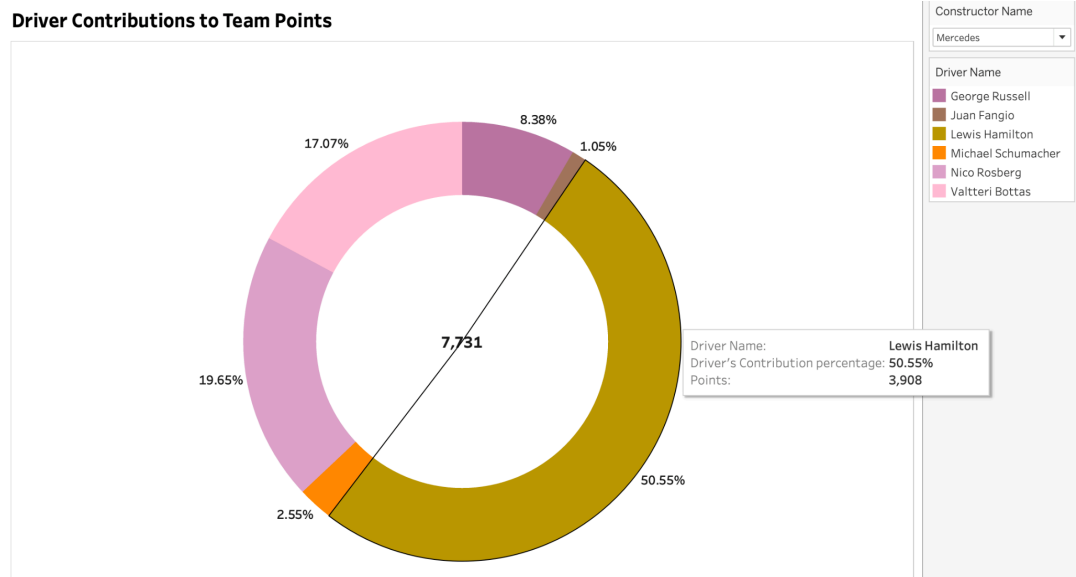
For example, Teams like Mercedes appear at the top 4 positions with nearly 1250 wins, showcasing the legacy and enduring performance over decades.

Discoveries from visualization

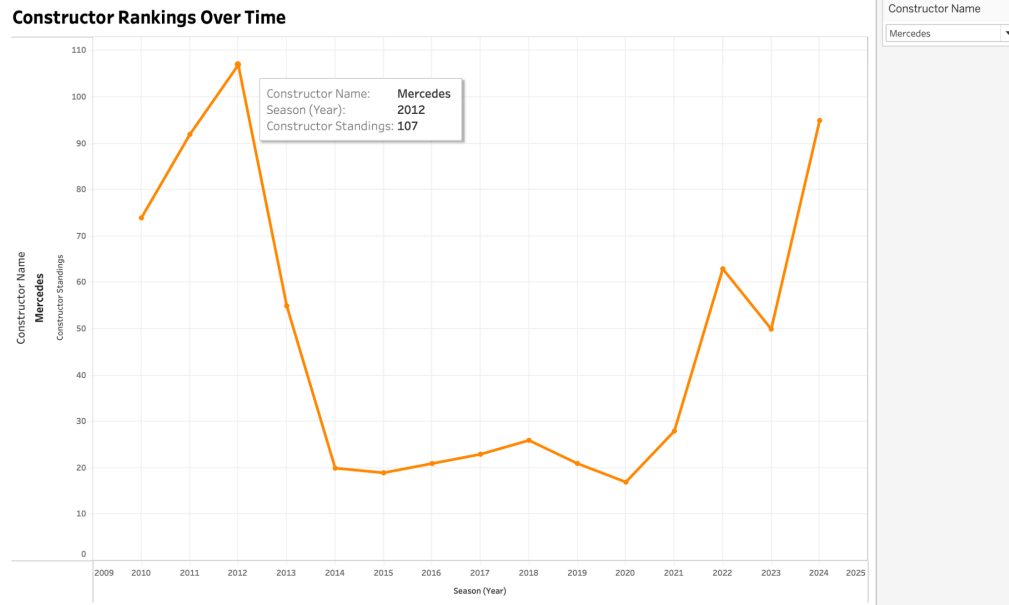
While exploring the data, two visualizations provided particularly valuable insights:

1. **Driver Contributions to Team Points:** This donut-shaped visualization unveiled clear disparities in how much each driver contributed to a constructor's total success. In the case of Mercedes, for instance, Lewis

Hamilton accounted for more than 50% of the team's cumulative points, revealing how heavily the team leaned on a single driver for consistent results. This kind of insight helps assess driver-team dependency and highlights standout performers across different eras.



- 2. Constructor Rankings Over Time:** The line chart mapping team standings year by year revealed distinct performance trajectories. While teams like Red Bull and Mercedes displayed sustained dominance, others experienced sudden drops or brief spikes in ranking. These fluctuations provide strategic insight into constructor evolution, investment impact, and how rule changes or driver line-ups may have influenced performance over time.



Challenges and Solutions

Developing the Constructor Success dashboard involved several real-world hurdles, especially in data integration and visualization accuracy. A **major challenge** was dealing with inconsistencies in raw CSV data, fields with \N placeholders, mismatched keys, and missing relationships across tables. This disrupted early join operations and made the Tableau model initially disjointed. Another common issue was Tableau's “row-level calculations can’t reference different upstream base tables” error, which blocked the use of combined filters like constructor and driver across different sheets. Designing KPIs such as “best performing driver” also posed a challenge due to the need for dynamic aggregation and context-sensitive filtering. Additionally, ensuring interactivity between filters, so the driver options update based on the selected constructor, required several calculated field experiments and level-of-detail expressions. Lastly, maintaining a clean, uncluttered visual layout while ensuring insights weren’t lost in simplicity was a constant balancing act.

To address these challenges, several solutions were taken into consideration,

1. For the data inconsistencies, early preprocessing in Python using Pandas and NumPy was essential. Duplicates were dropped, irrelevant columns were removed, and \N values were systematically replaced with "Unknown" to ensure Tableau could parse them properly.
2. In Tableau, relationships were restructured based on shared foreign keys (e.g., raceId, constructorId) instead of traditional joins to avoid unnecessary data duplication.
3. When KPI fields failed due to table conflicts, FIXED LOD expressions were used to isolate aggregate values within a constructor scope. For filtering logic between drivers and constructors, context filters were applied to enforce execution order.
4. The most effective workaround was restructuring calculated fields to be field-level rather than row-level, where possible. And from a UI perspective, a deliberate chart selection strategy like switching from bump to line charts or from bars to donuts helped declutter visuals while maintaining narrative clarity. Each challenge sharpened the focus of the dashboard and led to smarter, more stable visual storytelling.

Adjustments from the original plan

Initially, the plan was to use Tableau Prep Builder for data cleaning and table joins, assuming its visual interface would simplify the workflow. However, as the

dataset required more detailed preprocessing, such as handling of null values, reformatting date and time fields, and dropping irrelevant attributes, the process was shifted to Python, leveraging Pandas and NumPy for greater control and flexibility. Once the data was cleaned and structured, the joins were handled directly within Tableau Desktop using its data source modeling features, which allowed for smoother integration and easier field-level calculations. In addition to the tool shift, the project scope evolved as well, what began as a focus on tracking wins and standings grew into a broader analysis of driver contributions and historical team performance, ultimately leading to a more insightful and storytelling-oriented dashboard.

3. Race and Circuit Overview

This dashboard explores the spatial and temporal distribution of Formula 1 races, focusing on circuits, countries, and seasonal trends from the dataset. The underlying data is sourced from two primary CSV files: `races.csv` and `circuits.csv`.

Data Preparation and Integration:

This dashboard focuses on the importance and impact of qualifying performance in Formula 1. It examines how drivers and constructors fare in relation to their qualifying positions, nationalities, circuit characteristics, and consistency in position changes.

Data Integration and Modeling:

The core data source is `races.csv`, joined with four supporting datasets:

1. `circuits.csv` (joined via `circuitId`)
2. `qualifying.csv` (via `raceId` and `driverId`)

3. results.csv (via raceId and driverId)
4. status.csv (via statusId from results)

All joins were set up using Tableau's visual data model interface, primarily as left and inner joins to retain race records even when some related data was missing.

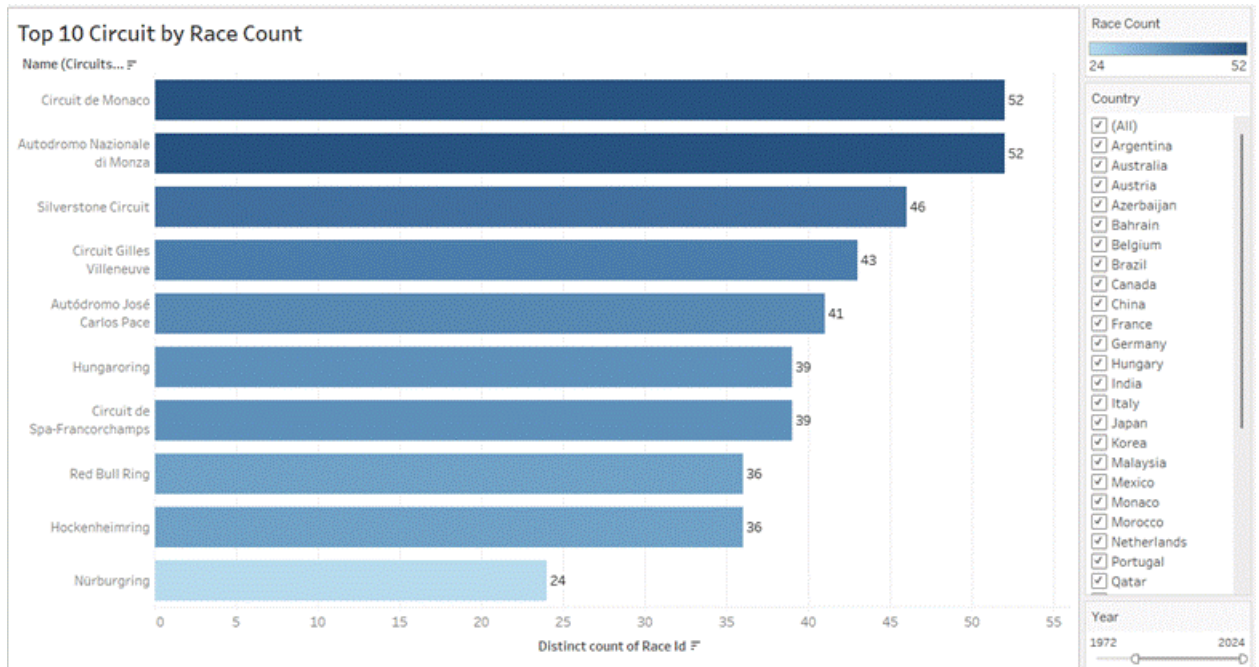
Fields were renamed for clarity and consistency. Geographic roles were assigned to circuit coordinates. Null values (such as missing sprint date, fp1 date or quali date) were handled using IFNULL() logic, often replacing them with "Unknown" or excluding them from specific charts to preserve accuracy.

Analysis of the Dashboard:

This dashboard comprises four primary visualizations that reveal trends and patterns in the geographical spread and frequency of Formula 1 races and circuits.

1. Top 10 Circuits by Race Count (Horizontal Bar Chart):

1. Shows which circuits have hosted the most F1 races.
2. Monza (52), Monaco (52), and Silverstone (46) top the list, indicating their historic and ongoing relevance.
3. Includes filters for Country and Year to analyze circuit dominance over time or by region.
4. This chart emphasizes the legacy of classic tracks and supports circuit-centric analysis.



2. Races by Country (Choropleth Map):

1. Maps the geographical distribution of races across the world.
2. Countries like the UK, Italy, Germany, and the USA are shaded darker, representing higher race frequency.
3. Users can filter the map by Year to examine the global expansion of F1 across decades.
4. Helpful for identifying emerging markets (e.g., Saudi Arabia, Qatar) and the Eurocentric origins of the sport.

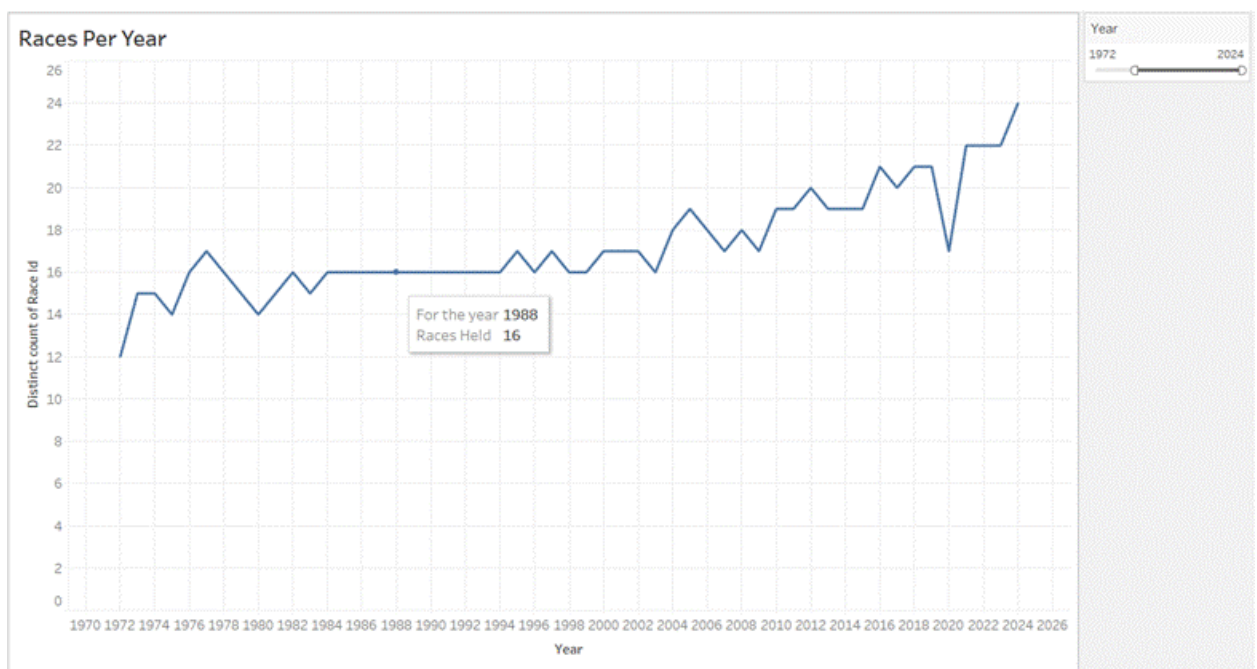
3. Percentage of DNS (Did Not Finish) Failure Reasons (Pie Chart):

1. Displays the percentage of failure reasons categorized by status such as: Engine, Collision, Accident, and multi-lap retirements like “+1 Lap” or “+2 Laps”.

2. Shows which mechanical or race incidents are most common causes of failures.
3. Filters: Status, Country, and Year enable granular exploration, e.g., whether collisions increased on certain street circuits.
4. Offers insights into how car reliability, driver error, or track risk factors impact race completion.

4. Average Number of Races Per Year (Line Chart):

1. Tracks how the number of races per season evolved from 1950 to 2024.
2. Clearly highlights the sport's growth trajectory, from fewer than 10 races in the early 1950s to over 23 in recent years.
3. The dip during COVID-impacted seasons like 2020 is visible, followed by a sharp resurgence.
4. Filters enable narrowing this growth by country, revealing how specific regions contributed to F1's expansion.



Discoveries from visualization:

1. Legacy Circuits Dominate: Monza, Monaco, and Silverstone consistently rank as the most used tracks, reinforcing their historical value.
2. Global Expansion: While the sport originated in Europe, recent years show a strong push into Asia, the Middle East, and the Americas.
3. Common Failure Reasons: Engine failures and collisions are among the most frequent DNF causes, particularly in high-speed or complex circuits.
4. Year-on-Year Growth: The chart of races per year demonstrates how F1 has evolved from a niche sport into a global entertainment giant.
5. Circuit-Country Relationship: Some countries (like Italy and Germany) feature multiple high-usage circuits, while others rely on a single venue.

Challenges and Solutions:

1. Selecting the correct data sources: The original dataset contained 14 CSV files, from which only 5 were used. Selecting the appropriate datasets demanded a careful evaluation of their relevance to the dashboard's goals, ensuring that only the most impactful sources were used.
2. Aggregations like COUNTD(Race ID) and COUNTD(Circuit Name) were used for distinct counts to avoid duplication from join expansions. Joining multiple datasets (like races.csv, results.csv) caused duplicate rows, inflating counts—resolved using COUNTD to ensure unique values for Race ID and Circuit Name.

3. Filters like Country or Year sometimes did not impact KPI sheets as expected, especially when the sheet's primary data source wasn't directly tied to the filter context; this was resolved by setting the filters to apply to all using this data source.

Adjustments from the original plan

1. Filter-Responsive KPIs Added: Initially, KPIs like “Average Races per Year” and “Total Circuit Count” were planned as static labels. Later, they were converted to dynamic metrics that update based on Year and Country filters for more meaningful interaction.
2. Simplified Layout for Clarity: Some visualizations (like average races or total circuits) were originally intended as detailed charts but were reduced to clean text KPIs for better readability and space optimization.
3. Map and Pie Chart Enhancements: Switched to filled maps instead of symbol maps to highlight countries by race frequency. Also changed the DNS reasons from a bar to a pie chart for better visual distribution.
4. Tooltip and Label Improvements: Added tooltips and percentage labels across visualizations to make insights more accessible without overwhelming the viewer.
5. Joined Additional Files Only Where Needed: Instead of joining all 14 available CSVs, selected only the relevant 5 based on dashboard objectives to keep the model clean and performant.

3. Dashboard Overview: F1 Qualifying Insights

This dashboard was designed to explore the relationship between qualifying performance and final race outcomes in Formula 1. It leverages data from five major CSV datasets: qualifying.cs33v, results.csv, races.csv, drivers.csv, and

constructors.csv. These files were integrated and analyzed using Tableau Desktop, where key calculated fields were created for in-depth performance metrics.

Key data preparation steps included:

1. Joining relevant datasets by driverId, raceId, and constructorId.
2. Cleaning and filtering null values in fields such as qualifying position and position order.
3. Converting qualifying session times (e.g., Q1, Q2) into numerical seconds for aggregation.
4. Creating calculated fields like Position Change, AvgQualifyingPosition, PoleToWinProbability, and fixed-LD values for ranking and KPIs.

Analysis of the Dashboard

The dashboard includes 5 visuals and 2 KPIs designed to answer strategic questions about race performance, driver improvement, constructor consistency, and nationality dominance.

KPI 1: Driver with Most Pole Positions

1. Purpose: Show the driver with the most pole positions in history.
2. Insight: Lewis Hamilton is highlighted as the leader. This KPI emphasizes this consistent one-lap performance across eras.
3. Real-world use: Reflects long-term dominance in qualifying sessions. Helps assess a driver's clean air advantage and track control.

KPI 2: Constructor with Best Avg Qualifying Position

1. Purpose: Identify which constructor has the lowest average qualifying position.

2. Insight: The dashboard shows “Simtek,” indicating either a data outlier or a case where limited but excellent performances were logged. Filter sensitivity may change this.
3. Real-world use: Offers team with the most consistent grid performance.

Pie Chart: Pole Positions by Nationality

1. Purpose: Visualize the global distribution of pole positions by country.
2. Insight: British and German drivers dominate, with notable contributions from Japanese, Austrian, and Colombian drivers. This reflects national motorsport investments.
3. Real-world use: Useful for comparing national driver development programs and predicting future talent pipelines.

Scatter Plot: Avg Qualifying Position vs Total Points Scored by Driver

1. Purpose: Evaluate how starting grid performance relates to championship points.
2. Insight: The trendline shows a negative correlation – better qualifying generally leads to more points. Some outliers with poor average starts but high points suggest strong racecraft.
3. Real-world use: Assists analysts in identifying under-utilized drivers who deliver more than their grid position would suggest.

Bar Chart: Driver’s Average Position Changes – Top 5

1. Purpose: Highlight drivers who regularly gained the most places from start to finish.
2. Insight: Jean-Marc Gounon and David Brabham lead the chart. These drivers gained on average 3–6 positions per race.
3. Real-world use: Valuable for identifying drivers with exceptional overtaking skills or consistent in-race performance boosts.

♦ **Bar Chart: Driver's Average Position Changes – Bottom 5**

1. Purpose: Identify drivers who consistently lost positions in races.
2. Insight: Roland Ratzenberger and Nigel Mansell are among those with the highest average position losses. This could reflect poor reliability, strategic mistakes, or decline in performance.
3. Real-world use: Teams could investigate causes of underperformance – was it technical failure, driver inconsistency, or team-level errors?

♦ **Bubble Chart: Pole-to-Win Probability by Circuit Over Years**

1. Purpose: Show which circuits offer the highest chances of converting pole position into a race win.
2. Insight: Larger and darker bubbles signify higher conversion. Circuits with multiple large bubbles above 0.9 win probability (like Monaco or Hungary) favor qualifying.
3. Real-world use: Critical for race engineers when optimizing car setups. Helps decide whether to emphasize qualifying pace or race strategy.

Interactive Filters and Dashboard Storytelling

The dashboard offers a range of filters that help users explore trends and derive personalized insights:

Year Filter (Slider)

1. Affects all visuals, especially bubble charts and scatter plot.
2. Story: Viewing 2010–2020 reveals Mercedes dominance; expanding to 1994 shows Schumacher's rise.

Constructor Filter

1. Impacts KPI for best qualifying constructor and scatter plot color encoding.

2. Story: Select “Ferrari” or “Red Bull” to see evolution across decades.

Nationality Filter

1. Affects pie chart and the driver-level KPIs.
2. Story: Selecting “Japanese” reveals role of lesser-known drivers and historical distribution of qualifying success.

These filters let users craft scenarios such as:

1. “Which circuits favor pole more in the last 5 years?”
2. “How do UK drivers compare to German drivers in pole performance?”
3. “Did driver X gain or lose positions more frequently after 2010?”

Discoveries from Visualization

1. Pole Conversion Circuits

Tracks like Monaco, Hungary, and Singapore show a high pole-to-win rate due to limited overtaking zones. The bubble chart confirms these circuits reward qualifying performance. In contrast, circuits like Spa and Baku show lower conversion, emphasizing race strategy and overtaking.

2. Race Gainers

The Top 5 Average Position Change bar chart shows drivers like Jean-Marc Gounon and David Brabham consistently gaining 5+ positions. This suggests exceptional race pace or ability to recover from poor grid starts.

3. Drivers Losing Out

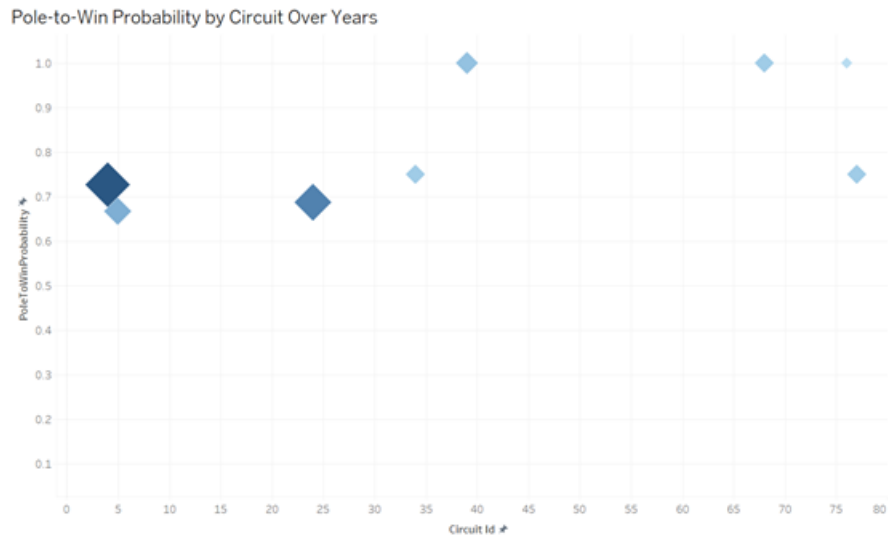
Drivers such as Nigel Mansell and Roland Ratzenberger appear in the Bottom

5, regularly dropping positions. This points to potential reliability issues, poor strategy, or race-day inconsistencies.



4. Constructors Vary by Era

When filtered by year and constructor, the line chart reveals shifts in dominance: Ferrari in the early 2000s, Red Bull in the early 2010s, and Mercedes during the hybrid era. These trends align with technical innovation and regulation changes.



5. Outliers in Scatter Plot

The scatter plot shows that although drivers with better average qualifying positions tend to score more points, some outliers (like Pérez or Button) still deliver high points from mid-grid starts — highlighting strategic excellence.

Challenges and Solutions

1. Originally intended to build the dashboard around pit stop strategies using the `pit_stops.csv` dataset. However, due to a lack of meaningful variation and limited visual diversity in the available pit stop data, the focus was shifted to qualifying performance, which offered richer insights and more dynamic visualizations.
2. KPI Logic Error: The “Best Constructor” showed unexpected results . Fixed by checking data granularity and excluding constructors with low entry count
3. Hidden Axis in Bubbles: Bubble chart had no X-axis by default. Solution: Convert circuit ID to continuous, and enable axis headers.
4. Color Conflicts in Bar Charts: Red and green bars clashed with filters. Used diverging color scale to distinguish gains from losses.

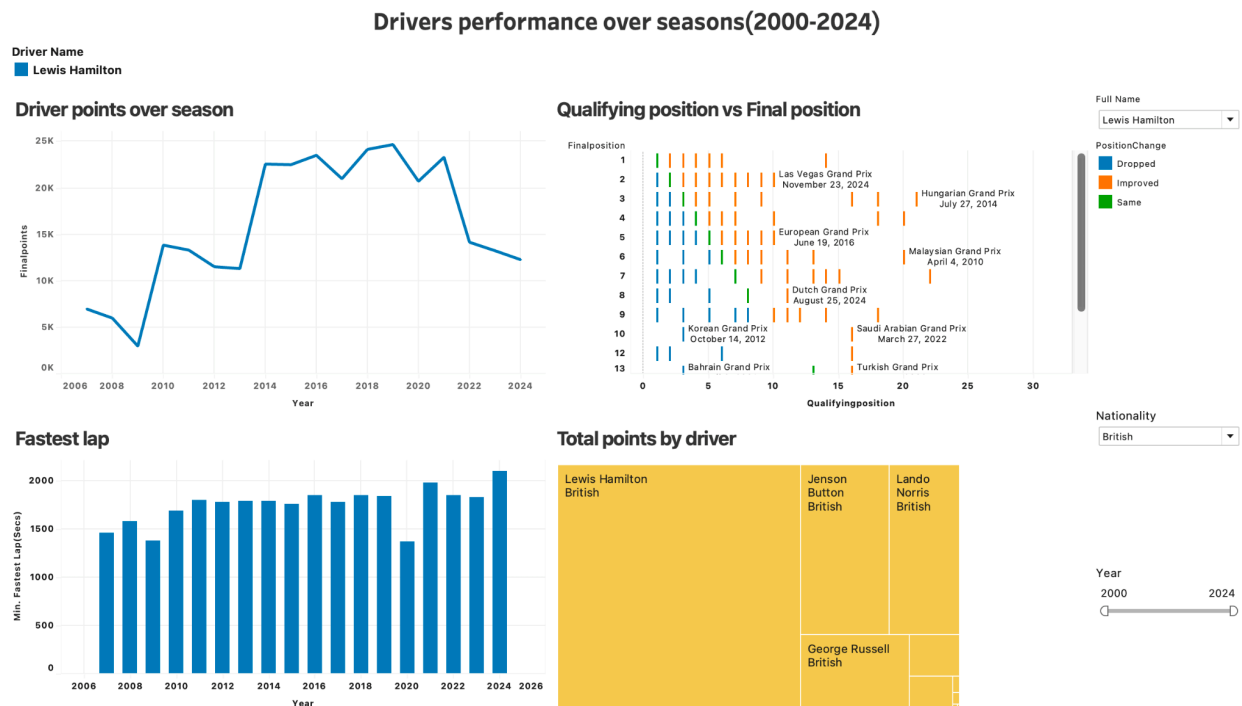
5. Disconnected Filters: Driver and constructor filters didn't update all views.
Resolved using dashboard filter actions and applying to all sheets.

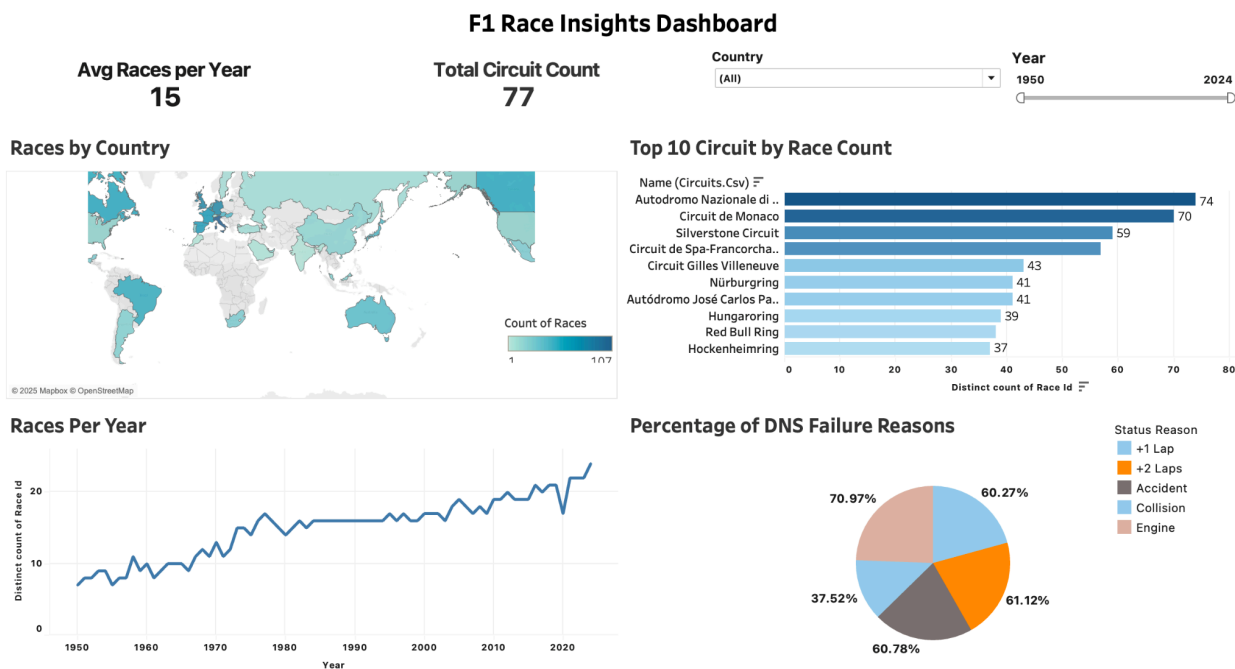
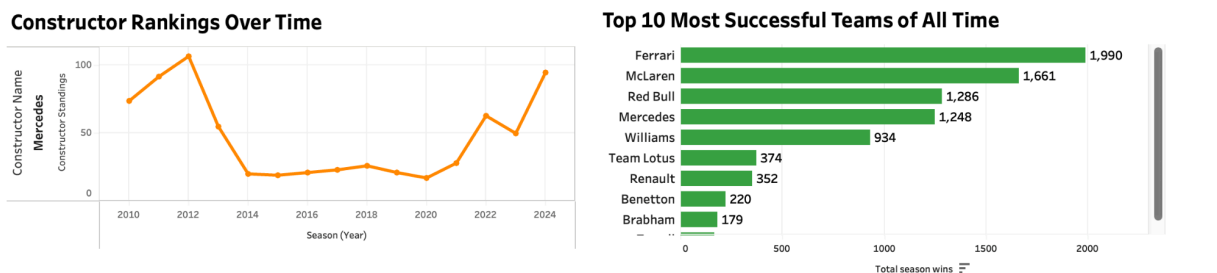
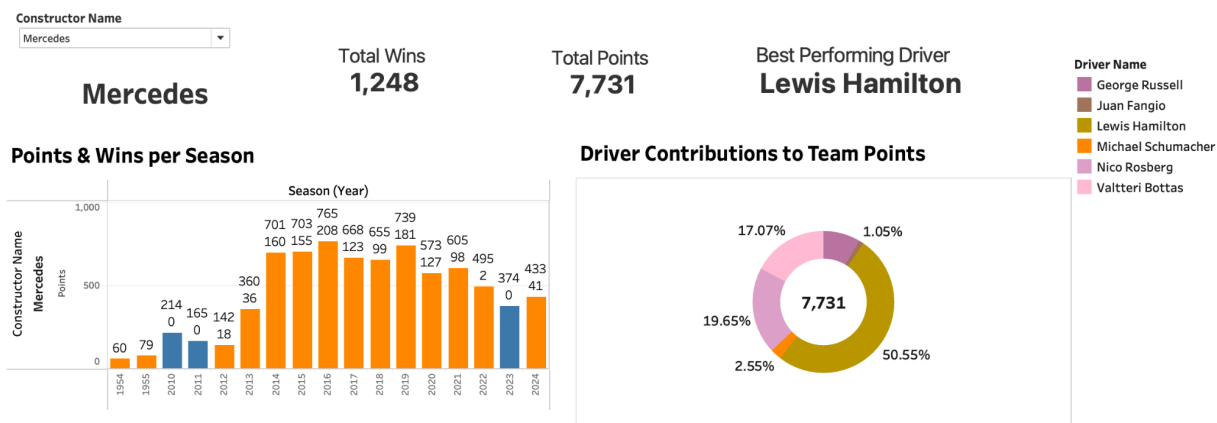
Adjustments from Original Plan

1. Initially planned for comparison across multiple drivers, but this cluttered charts. Shifted to top 5 / bottom 5 summaries for clarity.
2. Intended to include fastest qualifying laps, but dataset issues with formatting led to pivoting toward position change analysis instead.
3. Replaced the original “wins per season” bar chart with bubble chart for better insight into circuit-level race strategy.

This dashboard offers data-driven, filter-responsive insights into the strategic world of Formula 1 qualifying performance across drivers, teams, and race venues.

Final Product





F1 Qualifying Insights Dashboards

